

COMPUTER SCIENCE & IT

DIGITAL LOGIC



BOOLEAN THEOREMS AND GATES -1 Lecture : 05



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Recap of Previous Lecture

XOR & XNOR gate





Topics to be Covered

NAND & NOR gates

$$A \oplus B \oplus AB = (A+B)$$

$$A \oplus B \oplus (A+B) = A \cdot B$$

$$\begin{aligned} \bullet \quad A \oplus B \oplus \bar{A}B &= \overline{\bar{A} \oplus B \oplus \bar{A}B} \\ &= \overline{\bar{A} + B} \\ &= A \cdot \bar{B} \end{aligned}$$

$$\bullet \quad A \oplus \bar{B} \oplus \bar{A}B = \bar{A} \oplus B \oplus \bar{A}B = (\bar{A} + B)$$

$$\bullet \quad A \oplus B \oplus (\bar{A} + \bar{B}) = \bar{A} \oplus \bar{B} \oplus (\bar{A} + \bar{B}) = \bar{A} \cdot \bar{B}$$

$$\Rightarrow A \oplus B \oplus \overline{A \cdot B} = \overline{A \oplus B \oplus AB} = \overline{A+B}$$

$$\bullet \quad \bar{A} \oplus \bar{B} \oplus (\bar{A} + B)$$

$$= \overline{\bar{A} \oplus B \oplus (\bar{A} + B)}$$

$$= \overline{\bar{A} \cdot B}$$

$$= (A + \bar{B})$$

$$\bullet \quad AB \oplus BC = B \cdot [A \oplus C]$$

$$= \overline{A}B \cdot BC + AB \cdot \overline{BC}$$

$$= \overline{A}BC + AB\overline{C}$$

$$= B[\overline{A}C + A\overline{C}]$$

$$= B[A \oplus C]$$

$$AB \odot BC \neq B[A \odot C]$$

$$= \overline{A}B \cdot \overline{BC} + AB \cdot BC$$

$$= (\overline{A} + B) \cdot (\overline{B} + \overline{C}) + ABC$$

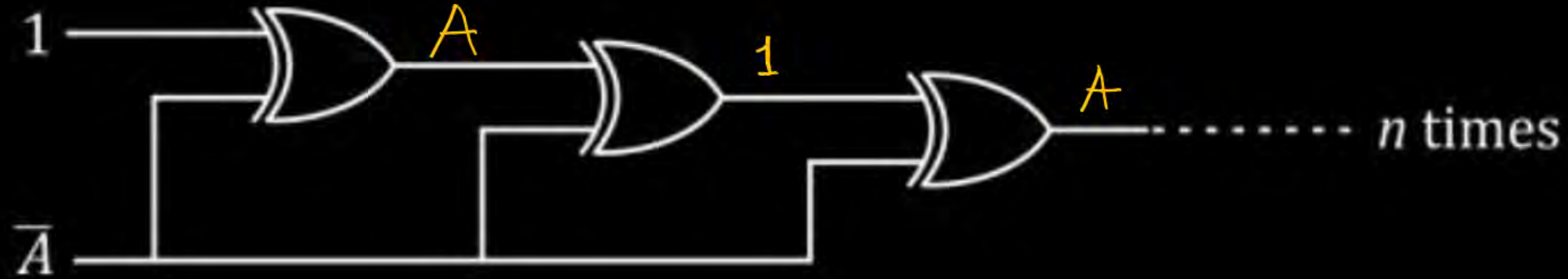
$$= \overline{B} + \overline{A} \cdot \overline{C} + ABC$$

$$= \overline{B} + AC + \overline{A}\overline{C}$$

$$= \overline{B} + A \odot C$$

$$\begin{aligned} &= \overline{AB \oplus BC} \\ &= \underline{\underline{B \cdot [A \oplus C]}} \\ &= \overline{B} + \overline{A \oplus C} \\ &= \overline{B} + A \odot C \end{aligned}$$

[Question]



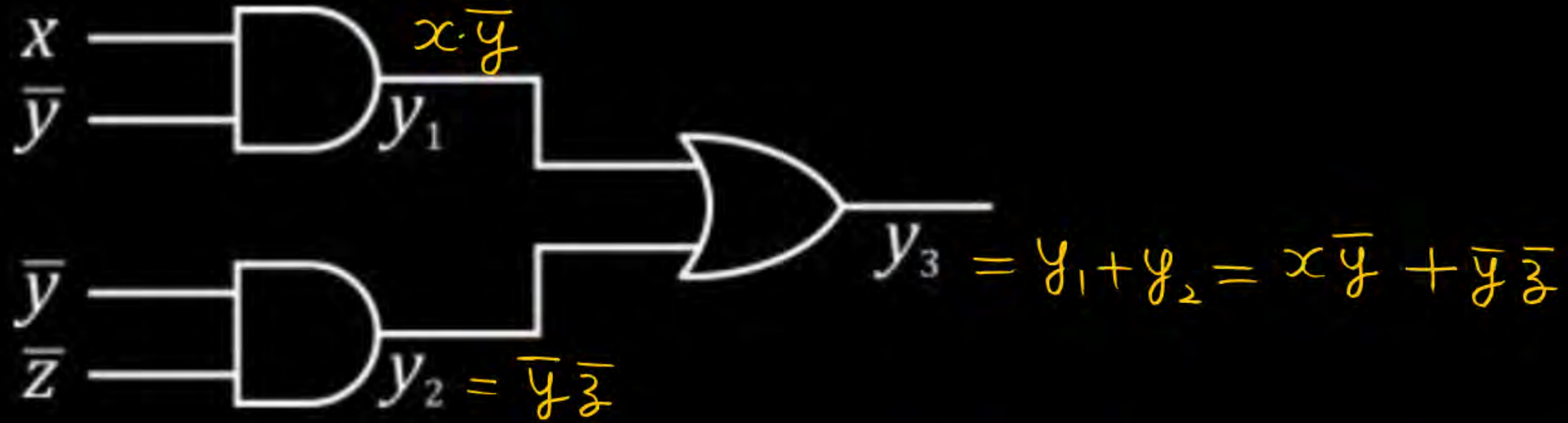
Then which of the following is true :

- | | | | |
|--------|--------------------------------|-----|--------------------------------------|
| (a) | Output is A if n is even ✗ | (b) | Output is $\bar{A}$ if n is even ✗ |
| (c) ✓✓ | Output is A if n is odd ✓✓ | (d) | Output is 1 if n is odd |

[Question]



A digital circuit is implemented as :



Then output $\underline{y} = \underline{y_1} \oplus \underline{y_2} \oplus \underline{y_3}$ will be

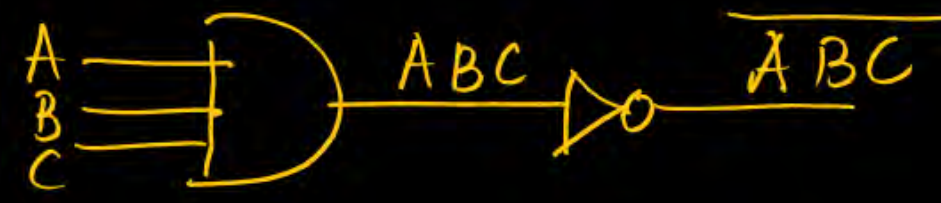
(a) $x\bar{y} + \bar{y}\bar{z}$ ✗

(b) $x \oplus y \oplus z$

(c) ✓ $x\bar{y}\bar{z}$

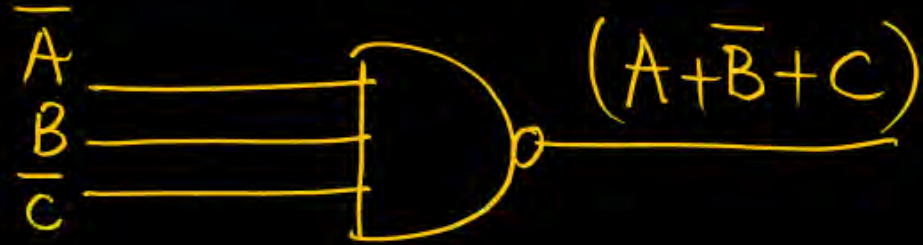
(d) $x + \bar{y} + \bar{z}$

$$\begin{aligned} y_1 \oplus y_2 \oplus y_3 &= y_1 \oplus y_2 \oplus (y_1 + y_2) \\ &= y_1 y_2 \\ &= x \bar{y} \bar{y} \bar{z} \\ &= x \bar{y} \bar{z} \end{aligned}$$

[NAND GATE] AND + NOT 



Representation :



	A	B	$y = \overline{A \cdot B}$
0	0	0	1
1	0	1	1
2	1	0	1
3	1	1	0

$$y(A, B) = \sum(0, 1, 2)$$

$$= \pi(3)$$

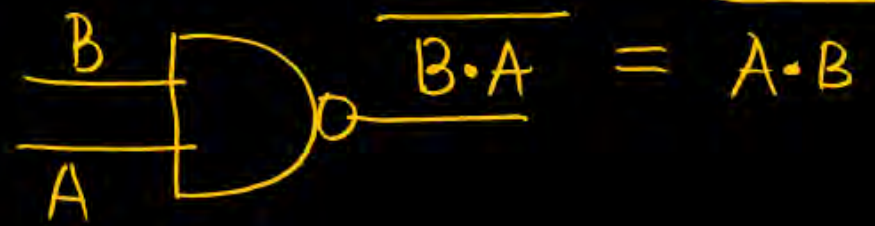
$$= (\overline{A} + \overline{B})$$

$$= \overline{A}\overline{B} + \overline{A}B + A\overline{B}$$

$$= \overline{A} + A\overline{B}$$

$$= (\overline{A} + \overline{B})$$

- Commutative Law :

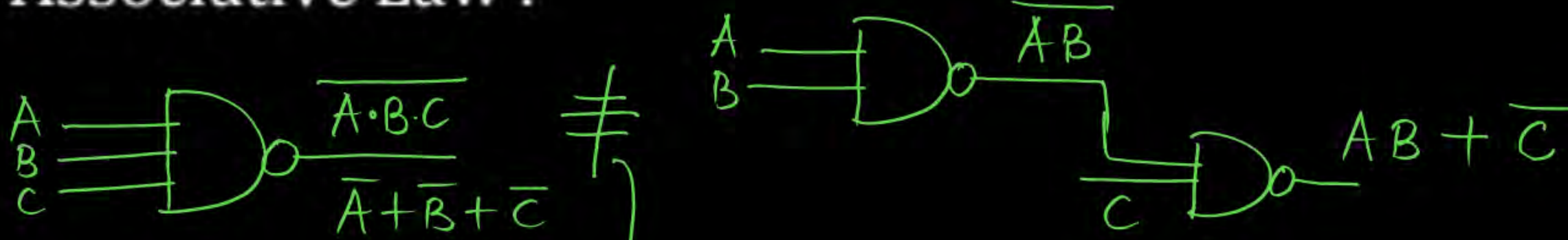


$$\overline{A \cdot B} = \overline{B \cdot A}$$

It holds commutative law.

→ position of variables is irrelevant.

- Associative Law :



It does not hold associative law.

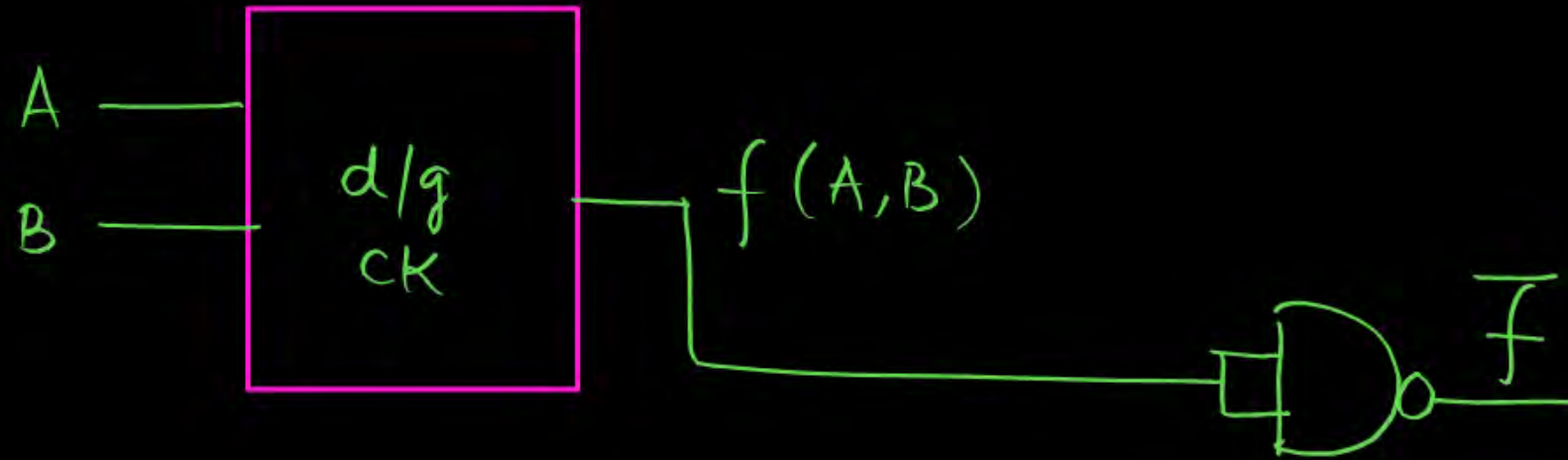
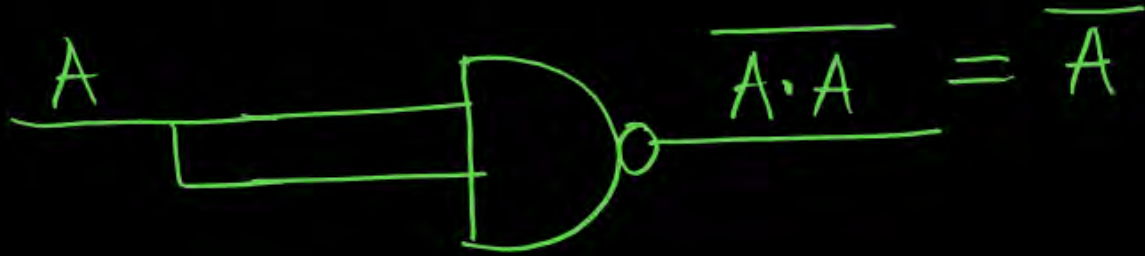
- IMP Points :

- If any one of the i/p line is at logic '0', then irrespective of other i/p lines, o/p has to be '1'.
- O/p will be '0' only if all the i/p's together are at logic '1'.

[NAND As an Universal Gate]

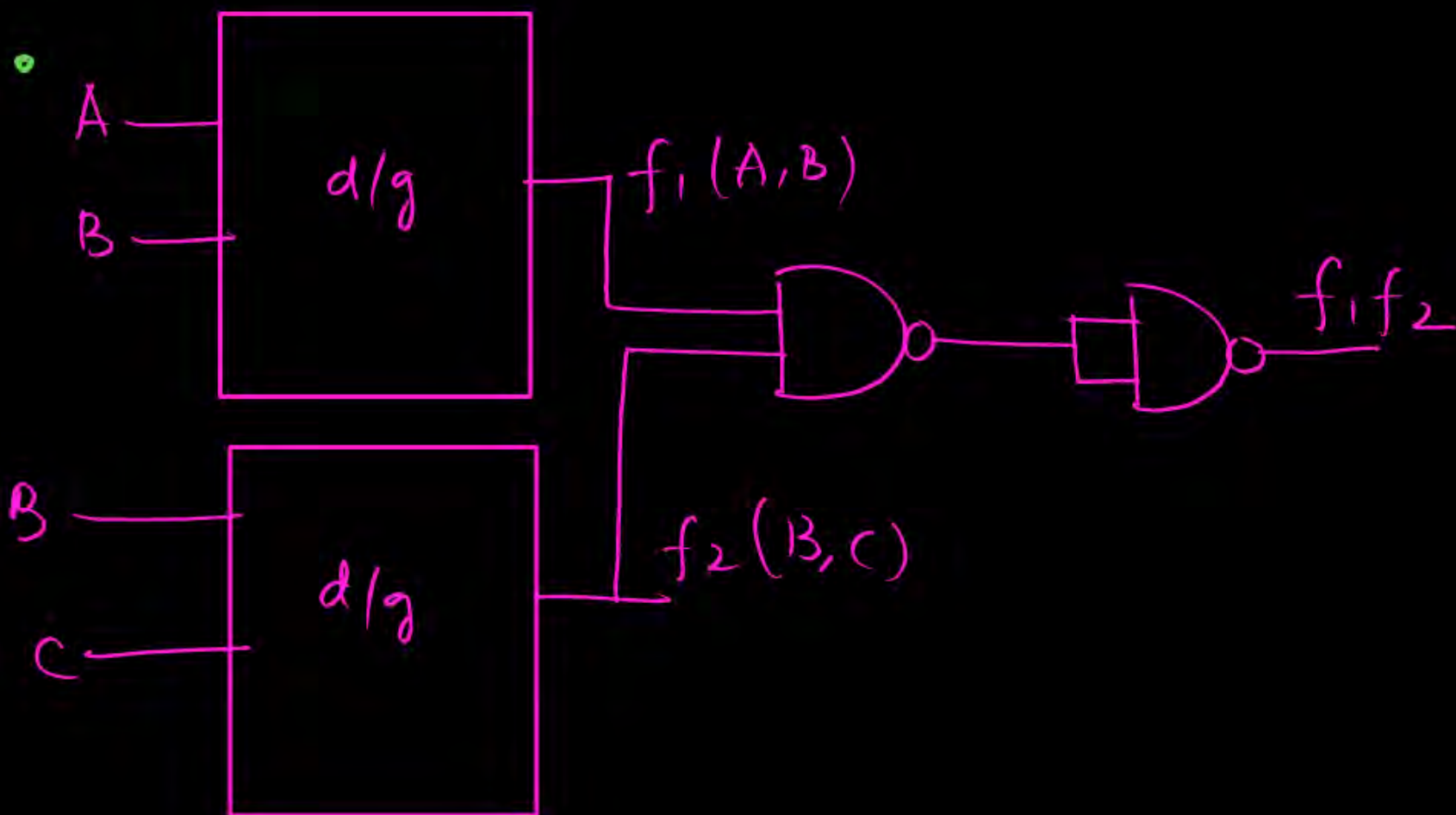
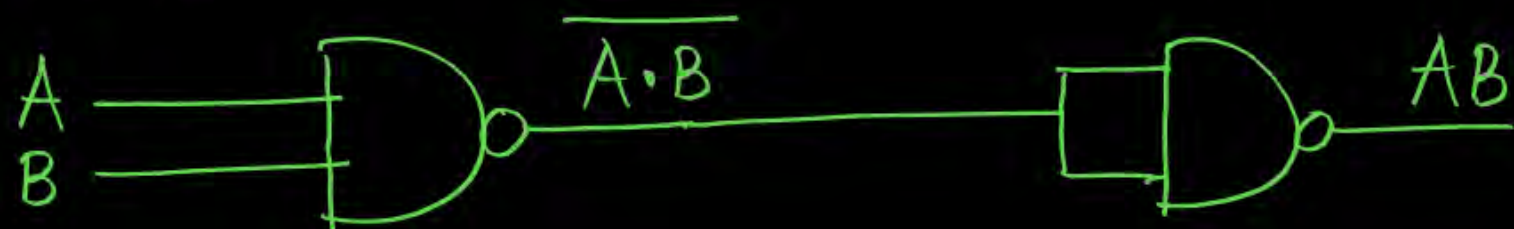


- NOT GATE :

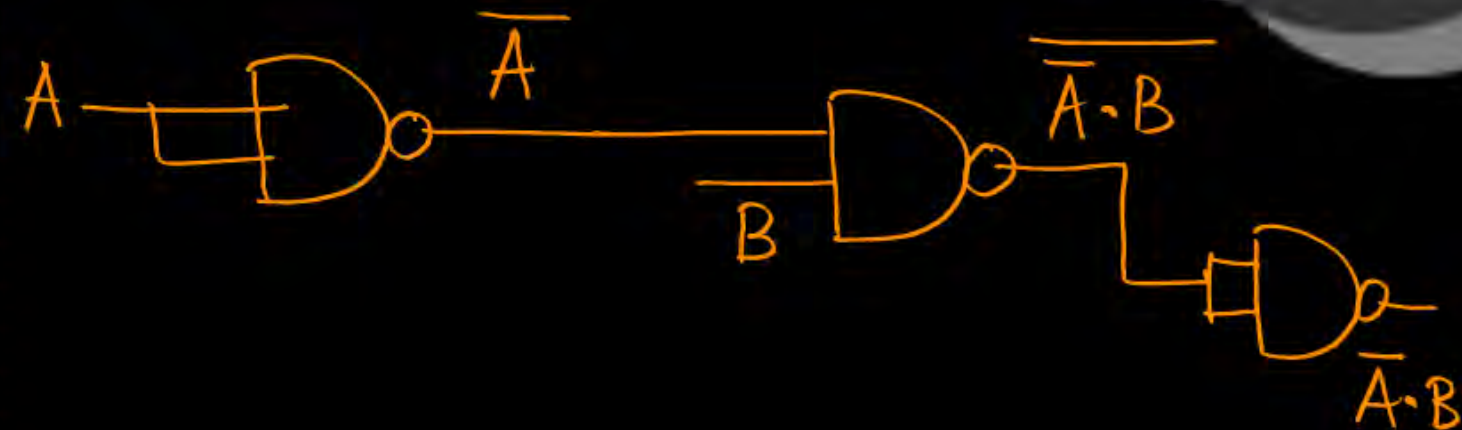


- AND GATE :

$$A \cdot B$$

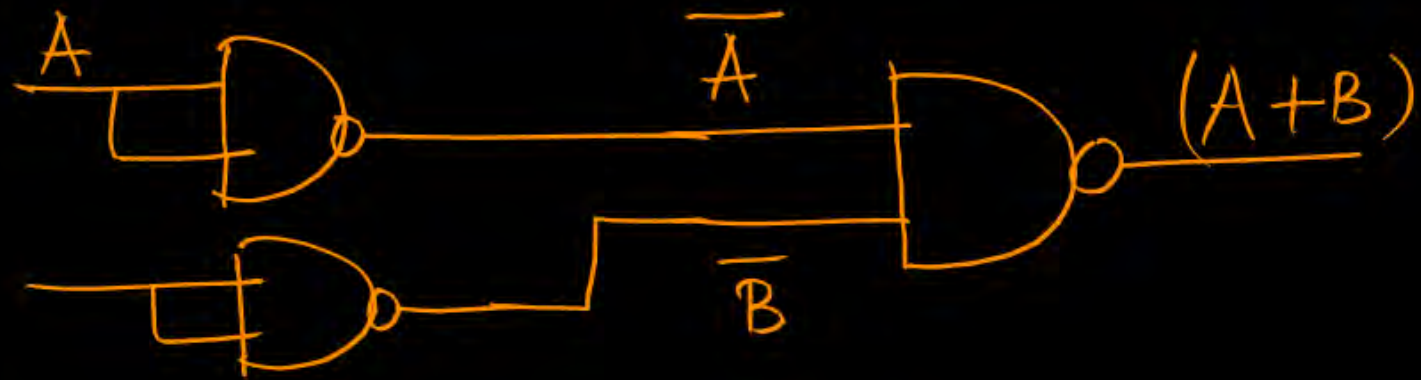


- $\overline{A} \cdot B$

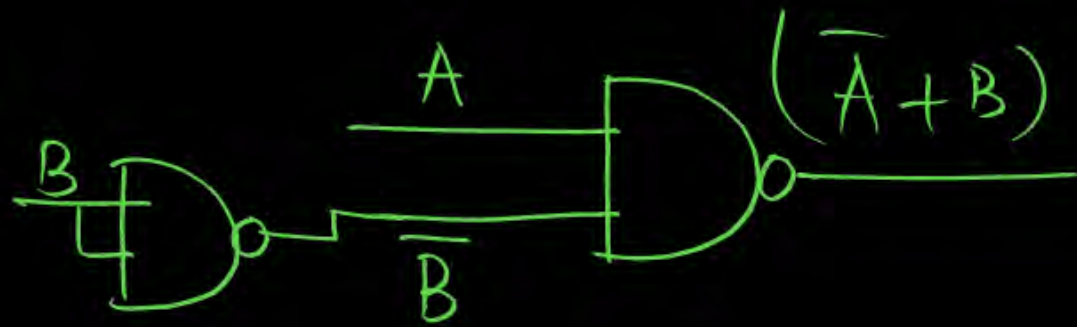


- OR GATE :

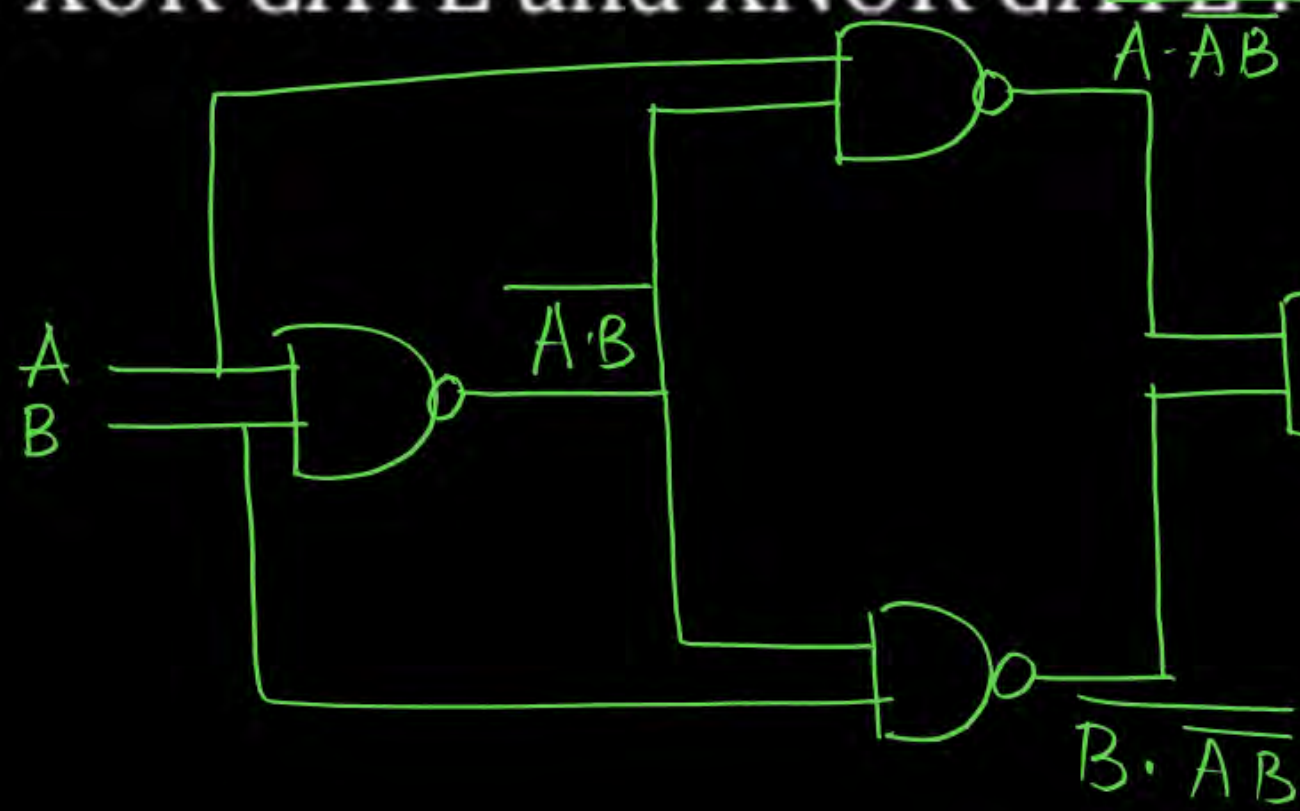
$$(A + B)$$



- $(\overline{A} + B)$



- XOR GATE and XNOR GATE:

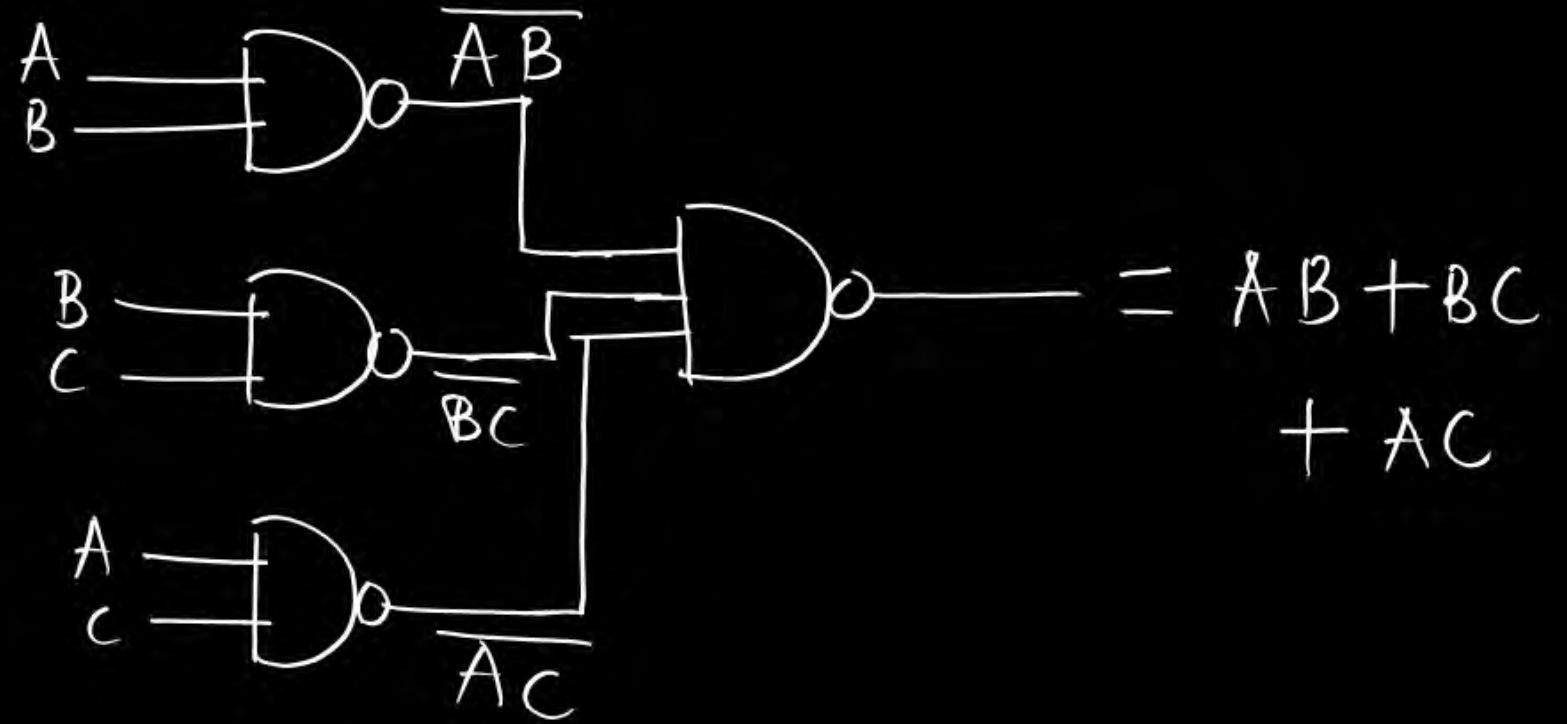
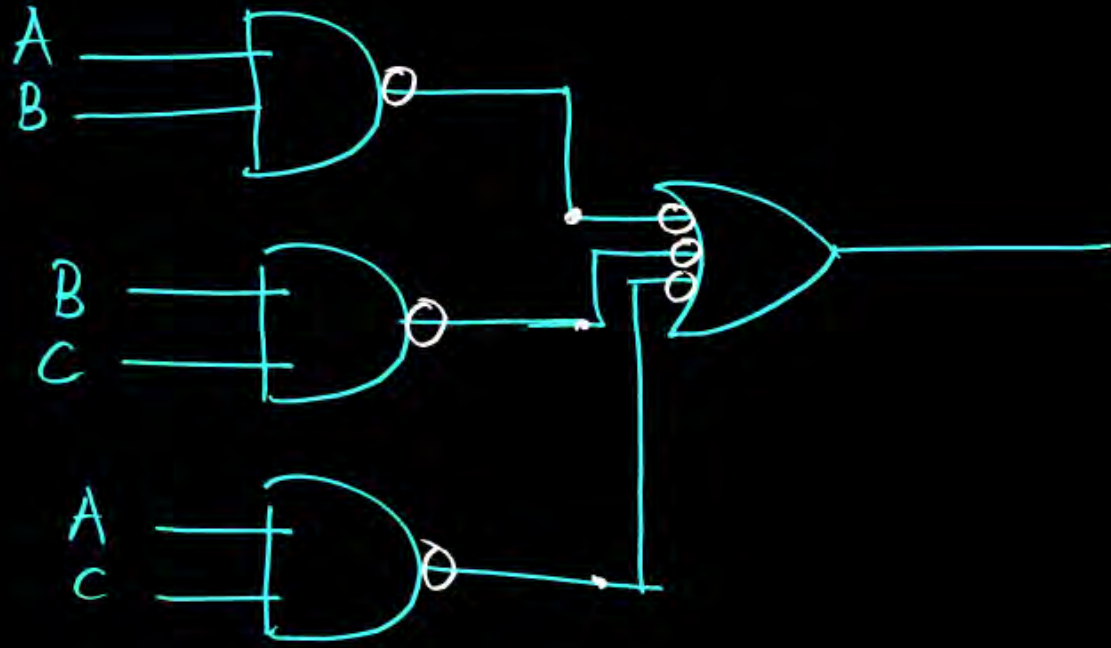


$$\begin{aligned}
 &= A \cdot \overline{A} B + B \cdot \overline{A} B = A \overline{B} + \overline{A} B \\
 &= \overline{A} B (A + B) \\
 &= (\overline{A} + \overline{B}) (A + B)
 \end{aligned}$$

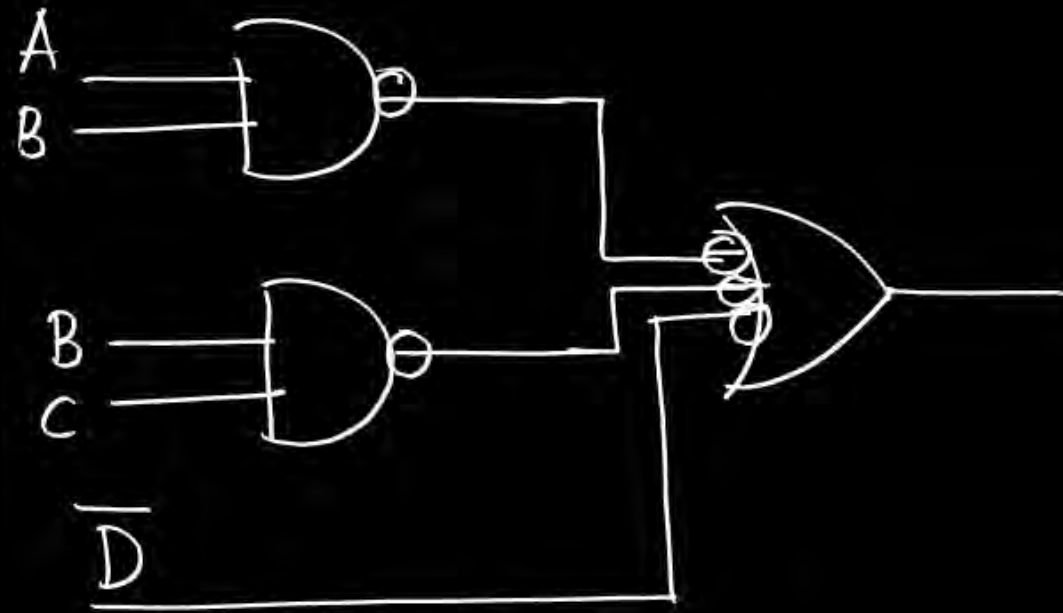
[2-Stage Implementation of SOP using NAND Gate]



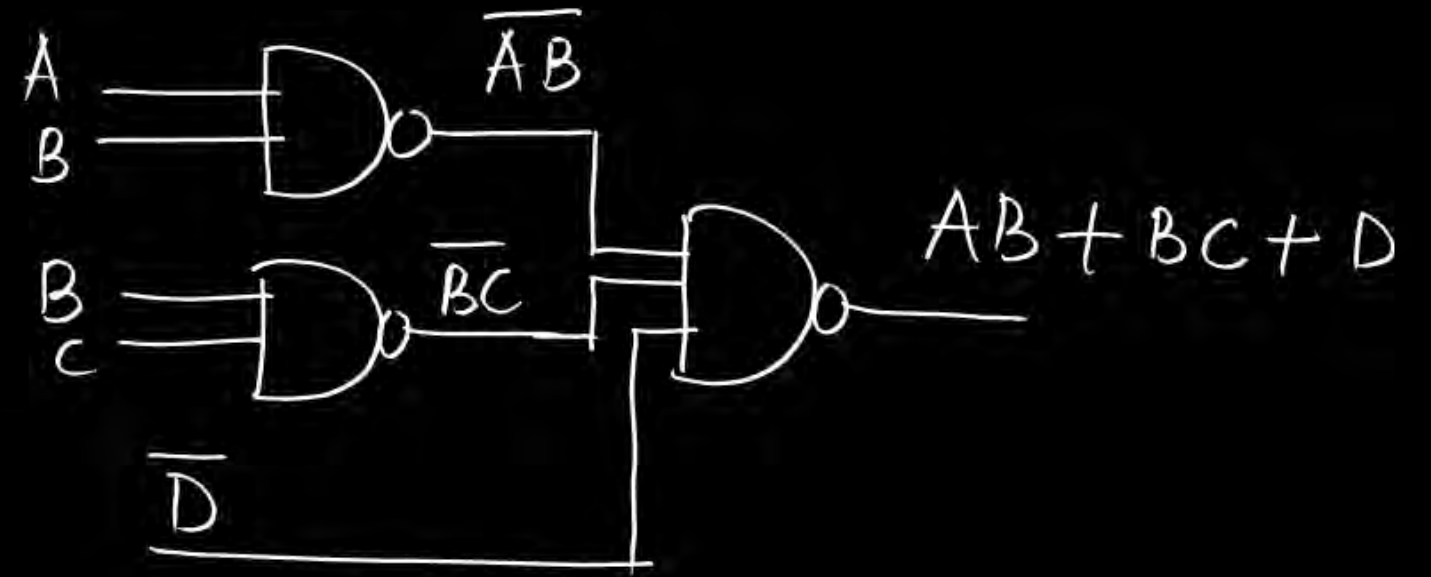
- $y = AB + BC + CA$



- $y = AB + BC + D$



$\equiv$

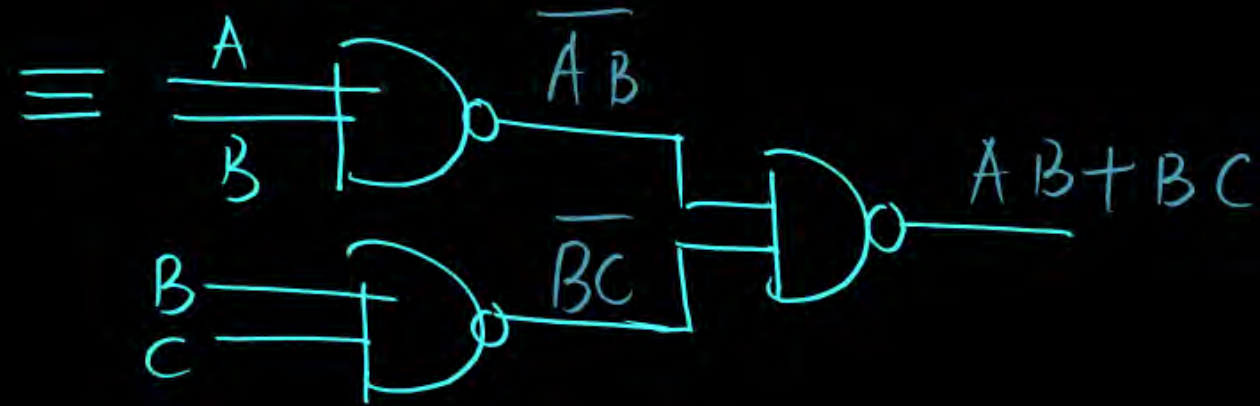
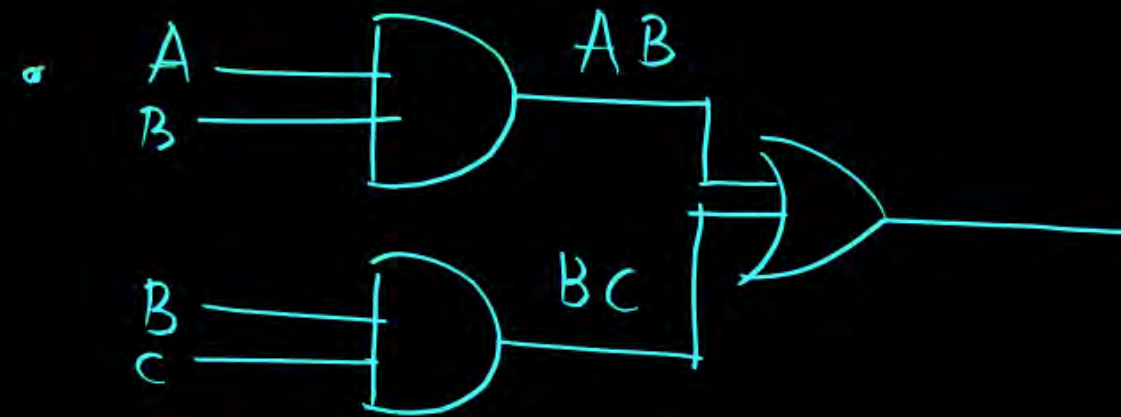


Question on minimum no. of NAND Gates

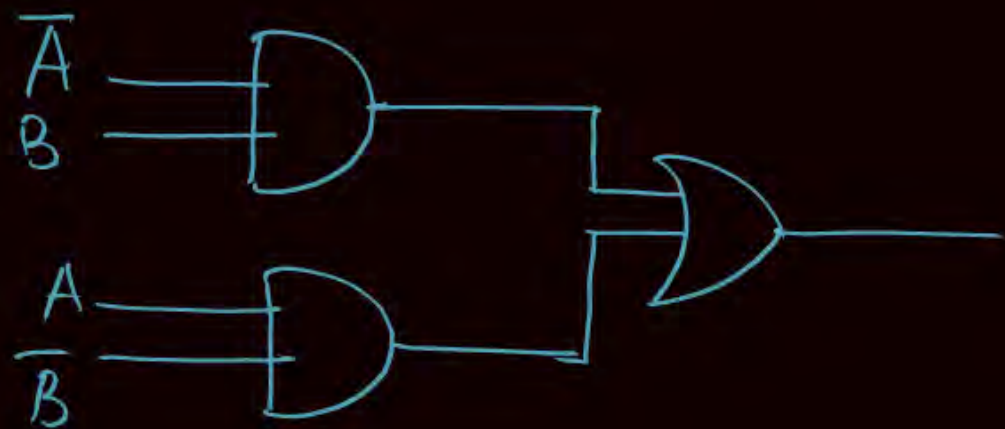
Question

$y = AB + BC$, to implement y , minimum no. of 2-input NAND gate required is 3.

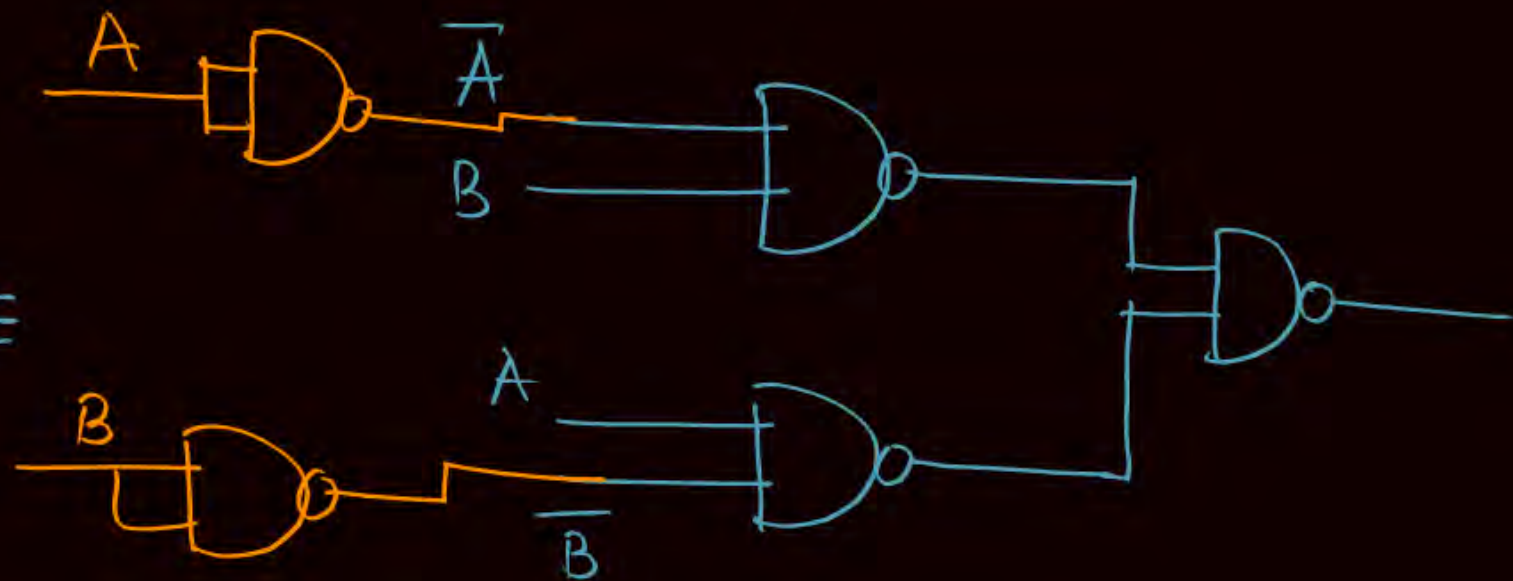
- $A \cdot B \rightarrow P \rightarrow 2 \text{ NAND}$
- $B \cdot C \rightarrow Q \rightarrow 2 \text{ NAND}$
- $P + Q \rightarrow 3 \text{ NAND}$



- $f(A, B) = \bar{A}B + A\bar{B}$



$\equiv$



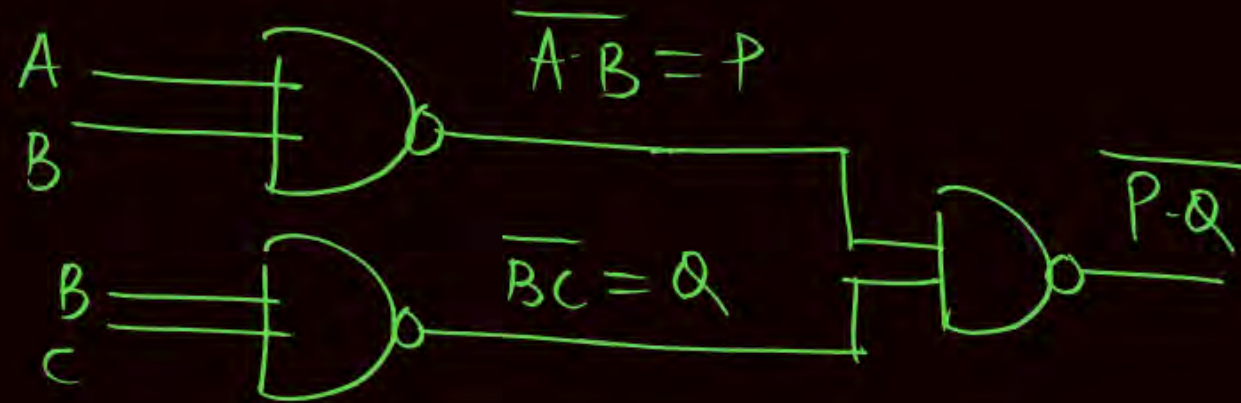
- $f(A, B, C) = AB + BC$

$$\overline{f} = \overline{AB + BC} = \overline{AB} \cdot \overline{BC}$$

$\downarrow$ $\searrow$
 $1N(P)$ $1N(Q)$

$$\overline{f} = P \cdot Q$$

$$f = \overline{P \cdot Q} \rightarrow 1N$$



$$A + B \rightarrow 3 \text{ NAND}$$

$$A \cdot B \rightarrow 2 \text{ NAND}$$

$$\overline{A \cdot B} \rightarrow 1N$$

- $f(A, B, C) = \overline{A}B + B\overline{C}$

$$\overline{f} = \overline{\overline{A}B} \cdot \overline{B\overline{C}}$$

$\downarrow$ $2N(P)$
 $\searrow$ $2N(Q)$



$$\overline{f} = P \cdot Q$$

$$f = \overline{P \cdot Q} \quad (1N)$$

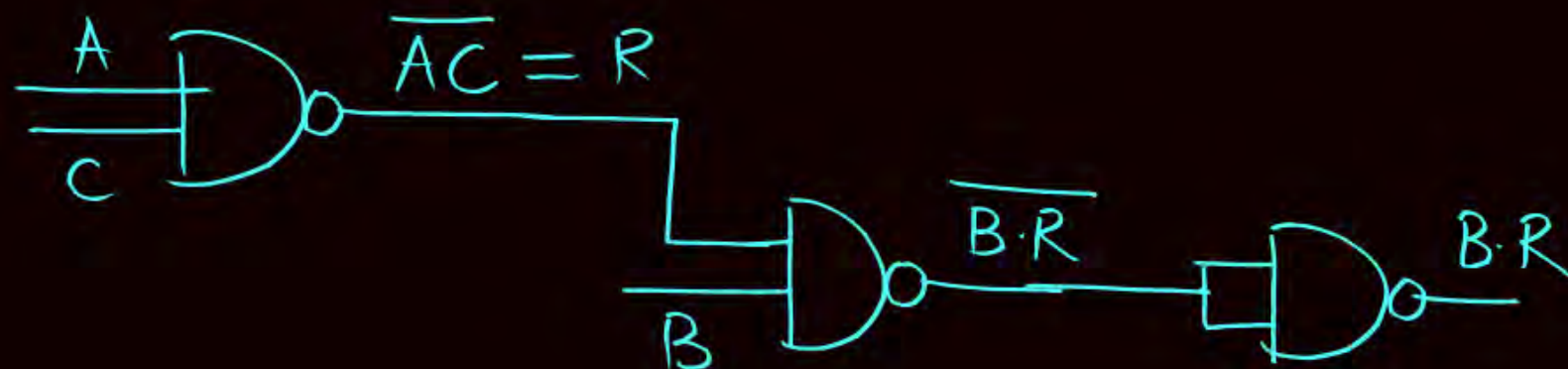
Total = 5

$$f = B(\overline{A} + \overline{C}) = B \cdot \overline{AC}$$

$\searrow$ $R(1N)$

$$= B \cdot R \quad [2N]$$

Total $\rightarrow 3$



$$\circ f(A, B) = \bar{A}B + A\bar{B} = (\bar{A} + \bar{B})(A + B)$$

$$= \overline{AB} (A + B) = (R) \cdot (A + B) = R \cdot A + R \cdot B$$

$$\downarrow$$

$$R(1N)$$

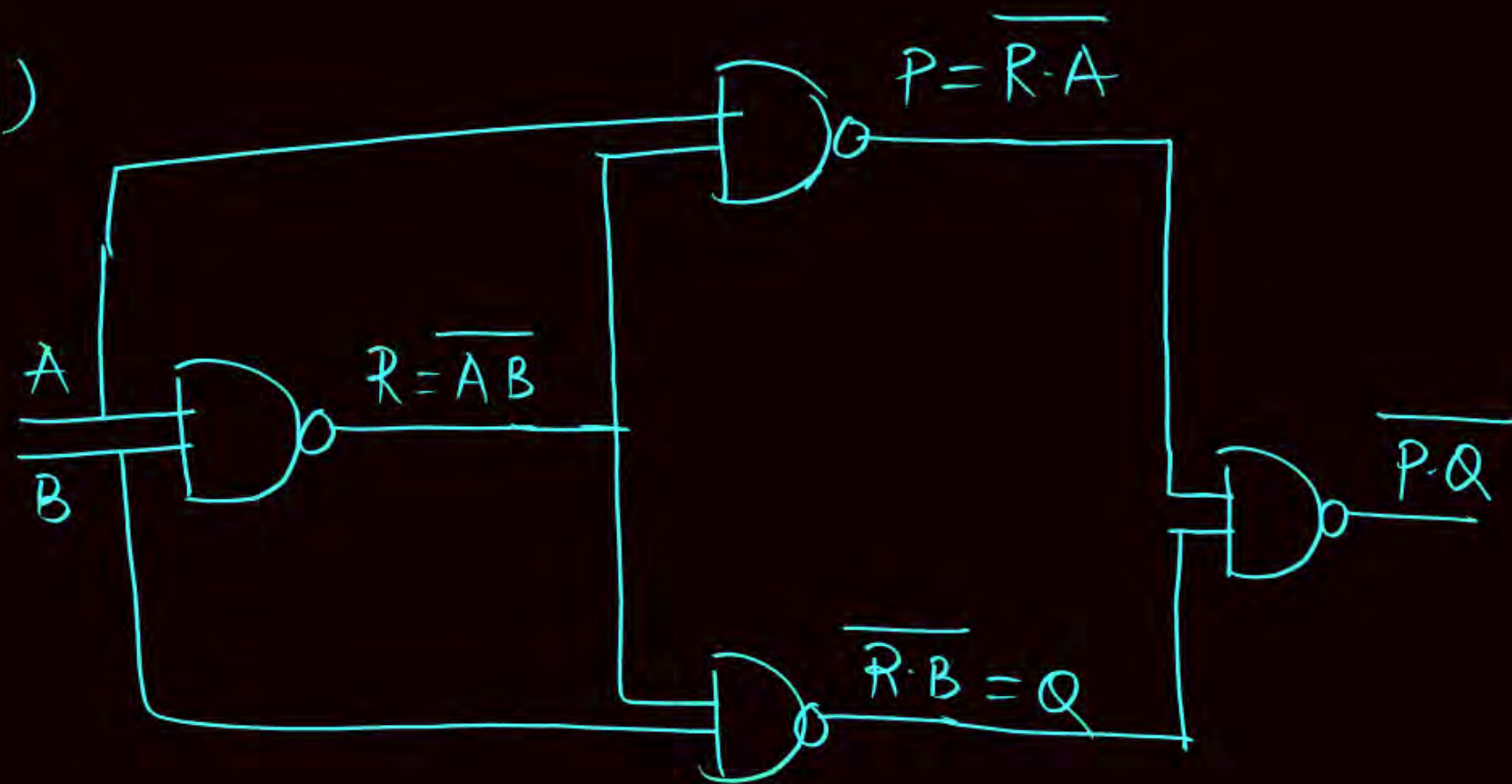
$$\bar{f} = \overline{RA} \cdot \overline{RB}$$

$$\downarrow \quad \quad \quad \searrow$$

$$IN(P) \quad \quad \quad IN(Q)$$

$$\bar{f} = P \cdot Q$$

$$f = \overline{P \cdot Q} \quad (1N)$$



$$\bullet f(A, B, C) = \overline{A}B + AC \quad \underline{\quad 4 \quad}$$

$$\begin{array}{cc} \overline{f} = & \overline{\overline{A}B} & \overline{AC} \\ & \downarrow & \downarrow \\ & 2N & 1N \\ & P & Q \end{array}$$

$$\overline{f} = P \cdot Q$$

$$f = \overline{P \cdot Q} \rightarrow 1N$$

• $f(A, B, C) = \overline{A \cdot B \cdot C}$ $\longrightarrow$ minimum no. of 2-C/P NAND required to implement f _____.

$$A \cdot B \longrightarrow P \text{ (2N)}$$

$$f = \overline{P \cdot C} \text{ (1N)}$$



• $f(A, B, C) = \bar{A}BC \rightarrow$ minimum no. of 2-i/p NAND 5

$$B \cdot C \rightarrow 2N(P)$$

$$f = \bar{A} \cdot P = Q \cdot P [2N]$$

$\downarrow$
IN (Q)

H.W.

find out minimum no. 2-i/p NAND gate require to implement functions given below:

Q.1 $f(A, B, C) = \overline{A}B + A\overline{C}$

Q.2 $f(A, B, C) = \overline{A}\overline{B} + A\overline{C} + B\overline{C}$

Q.3. $f(A, B, C) = AB + BC + AC$

Q.4. $f(A, B, C) = \overline{AB} + \overline{BC}$

Q.5. $f(A, B, C, D) = \overline{AB} + \overline{CD}$

Q.6. $f(A, B, C) = \overline{ABC}$



2 Minute Summary

→ XOR & XNOR
→ NAND

Thank you

GW
Soldiers !

